2.46 2nd Order DEs: Simple Harmonic Motion

Worked Solutions

1) a

$$x = 3\cos 2t - 4\sin 2t, \qquad \dot{x} = \frac{dx}{dt} = -6\sin 2t - 8\cos 2t.$$

1) b

$$\ddot{x} = \frac{d^2x}{dt^2} = -12\cos 2t + 16\sin 2t.$$

1) c

$$\ddot{x} = -\omega^2 x \implies -12\cos 2t + 16\sin 2t = -\omega^2(3\cos 2t - 4\sin 2t).$$

As $-12\cos 2t + 16\sin 2t = -4(3\cos 2t - 4\sin 2t)$, $\ddot{x} = -4x$, so $\omega^2 = 4$ and $\omega = 2$ (the angular frequency is taken to be positive).

2) a

$$x = A\cos 3t + B\sin 3t, \quad \dot{x} = -3A\sin 3t + 3B\cos 3t, \quad \ddot{x} = -9A\cos 3t - 9B\sin 3t = -9x.$$

Thus $\omega^2 = 9$ and $\omega = 3$.

2) b

At $t = 0$, $x = 2$ gives $2 = A\cos 0 + B\sin 0$, so $A = 2$. Also

$$-3 = \dot{x}(0) = -3(2)\sin 0 + 3B\cos 0 = 3B,$$

so $B = -1$. Hence $A = 2$ and $B = -1$.

3)

The auxiliary equation is $m^2 + \omega^2 = 0$, so $m^2 = -\omega^2$ and $m = \pm\omega i$. Hence the general solution is

$$x = A\cos\omega t + B\sin\omega t.$$

4)

The general solution is $x = A\cos\omega t + B\sin\omega t$. Substituting $x = x_0$ and $t = 0$ gives $x_0 = A$, so $x = x_0\cos\omega t + B\sin\omega t$. Differentiating,

$$\dot{x} = -\omega x_0\sin\omega t + B\omega\cos\omega t.$$

Substituting $\dot{x} = v_0$ and $t = 0$ gives $v_0 = B\omega$, so $B = \frac{v_0}{\omega}$. Therefore

$$x = x_0\cos\omega t + \frac{v_0}{\omega}\sin\omega t.$$

5) a

The auxiliary equation is $m^2 + 36 = 0$, so $m = \pm 6i$. Thus $x = A\cos 6t + B\sin 6t$. Substituting $x = 3$ at $t = 0$ gives $A = 3$. Then $x = 3\cos 6t + B\sin 6t$. At $t = \pi/12$, $x = 4$:

$$4 = 3\cos\frac{\pi}{2} + B\sin\frac{\pi}{2} = 0 + B,$$

so $B = 4$. Hence $x = 3\cos 6t + 4\sin 6t$.

5) b

The auxiliary equation is $m^2 + 400 = 0$, so $m = \pm 20i$. Thus $x = A\cos 20t + B\sin 20t$. At $t = 0$, $x = -5$, so $A = -5$. At $t = \frac{3\pi}{40}$, $x = 8$:

$$8 = -5\cos\frac{3\pi}{2} + B\sin\frac{3\pi}{2} = 0 - B,$$

so $B = -8$. Hence $x = -5\cos 20t - 8\sin 20t$.

5) c

The auxiliary equation is $m^2 + 2 = 0$, so $m = \pm\sqrt{2}i$. Thus $x = A\cos(\sqrt{2}t) + B\sin(\sqrt{2}t)$. Since $x(0) = 1$, $A = 1$. Differentiating,

$$\dot{x} = -\sqrt{2}\sin(\sqrt{2}t) + \sqrt{2}B\cos(\sqrt{2}t).$$

Using $\dot{x}(0) = \frac{1}{3}$ gives $\sqrt{2}B = \frac{1}{3}$, so $B = \frac{\sqrt{2}}{6}$. Therefore

$$x = \cos(\sqrt{2}t) + \frac{\sqrt{2}}{6}\sin(\sqrt{2}t).$$

5) d

The auxiliary equation is $25m^2 + 1 = 0$, so $m = \pm\frac{1}{5}i$. Thus $x = A\cos(t/5) + B\sin(t/5)$. Since $x(0) = \frac{1}{20}$, $A = \frac{1}{20}$. Then

$$\dot{x} = -\frac{1}{100}\sin(t/5) + \frac{1}{5}B\cos(t/5).$$

At $t = 10\pi$, $\dot{x} = \frac{1}{100}$:

$$\frac{1}{100} = -\frac{1}{100}\sin 0 + \frac{1}{5}B\cos 2\pi,$$

so $\frac{1}{5}B = \frac{1}{100}$, and $B = \frac{1}{20}$. Hence $x = \frac{1}{20}\cos(t/5) + \frac{1}{20}\sin(t/5)$.

6) a

$$x = 3\cos t + 4\sin t = 5\sin\left(t + \arctan\frac{3}{4}\right).$$

6) b

$$x = 4\cos 2t - 5\sin 2t = \sqrt{41}\cos\left(2t + \arctan\frac{5}{4}\right).$$

6) c

$$x = 4\sin 3t - 2\cos 3t = 2\sqrt{5}\sin(3t - \arctan 2).$$

7) a

The auxiliary equation is $m^2 + 9 = 0$, so $m = \pm 3i$ and $x = A\cos 3t + B\sin 3t$. Using $x(0) = 2$ gives $A = 2$, so $x = 2\cos 3t + B\sin 3t$. Then $\dot{x} = -6\sin 3t + 3B\cos 3t$. At $t = \frac{\pi}{3}$, $\dot{x} = 6$:

$$6 = -6\sin\pi + 3B\cos\pi = -3B,$$

so $B = -2$. Thus $x = 2\cos 3t - 2\sin 3t$.

7) b

$$x = 2\cos 3t - 2\sin 3t = 2\sqrt{2}\cos\left(3t + \frac{\pi}{4}\right).$$

8) a

$$\begin{aligned} x\left(t+\frac{2\pi}{\omega}\right) &= A\cos\left(\omega\left(t+\frac{2\pi}{\omega}\right)\right)+B\sin\left(\omega\left(t+\frac{2\pi}{\omega}\right)\right) \\ &= A\cos(\omega t+2\pi)+B\sin(\omega t+2\pi) \\ &= A(\cos\omega t\cos 2\pi-\sin\omega t\sin 2\pi)+B(\sin\omega t\cos 2\pi+\cos\omega t\sin 2\pi) \\ &= A\cos\omega t+B\sin\omega t = x(t), \end{aligned}$$

as required.

8) b

The period is $T=\frac{2\pi}{\omega}$.

9)

$$\begin{aligned} A\cos\omega t+B\sin\omega t &\equiv R\cos(\omega t-\phi) \\ &\equiv R(\cos\omega t\cos\phi+\sin\omega t\sin\phi) \\ &\equiv R\cos\phi\cos\omega t+R\sin\phi\sin\omega t. \end{aligned}$$

Comparing coefficients, $A=R\cos\phi$ and $B=R\sin\phi$, so $R=\sqrt{A^2+B^2}$ and $\tan\phi=\frac{B}{A}$. Thus

$$\phi=\arctan\frac{B}{A}, \qquad A\cos\omega t+B\sin\omega t\equiv\sqrt{A^2+B^2}\cos\left(\omega t-\arctan\frac{B}{A}\right).$$

10) a

The acceleration $\ddot{x}$ is proportional to the displacement x with constant $-\omega^2$. Because of the negative sign, the acceleration will always be pointing in the opposite direction to the displacement.

10) b(i)

If $x>0$, then $\ddot{x}=-\omega^2x<0$.

10) b(ii)

If $x<0$, then $\ddot{x}=-\omega^2x>0$.

10) b(iii)

The acceleration always points towards $x=0$, so $x=0$ is an equilibrium point and the motion is restoring towards equilibrium.

11) a

The acceleration $\ddot{x}$ is proportional to the displacement x with constant -7. Because of the negative sign, the acceleration will always be pointing in the opposite direction to the displacement.

11) b

At equilibrium, acceleration must be zero. Substitute $x=0$:

$$\ddot{x}=-7\times 0=0.$$

So if the particle is at $x=0$ and at rest, it will remain there. Hence $x=0$ is the equilibrium position.

11) c

If the particle passes through $x=0$ with positive velocity, it is moving in the positive direction. Immediately after passing through, $x>0$ and so $\ddot{x}=-7x<0$. The acceleration is negative, so the particle begins to slow down as it moves in the positive direction.

12) a

Since $x > 0$, $\ddot{x} = -\omega^2 x < 0$, so the acceleration is negative.

12) b

If $x > 0$ and $\dot{x} < 0$, the particle is moving in the negative direction back towards $x = 0$. So the particle is moving towards equilibrium.

12) c

As $\dot{x} < 0$ and $\ddot{x} < 0$, the particle is accelerating in the negative direction, so the speed is increasing (until it hits its maximum speed at the equilibrium point).

13) a

$$x(t) = R\cos(\omega t - \phi), \qquad \dot{x}(t) = -R\omega\sin(\omega t - \phi).$$

13) b

Speed $= |\dot{x}(t)| = |-R\omega\sin(\omega t - \phi)|$. The maximum speed is $R\omega$ as the maximum value of $\sin(\omega t - \phi)$ is 1.

13) c

Turning points occur when $\dot{x} = 0$:

$$-R\omega\sin(\omega t - \phi) = 0 \implies \omega t - \phi = k\pi,$$

where k is an integer. Then $x = R\cos k\pi = \pm R$. Therefore $x = \pm R$ when $t = \frac{\phi + k\pi}{\omega}$, where k is an integer.

14) a

Substitute $t = \frac{\phi}{\omega}$ into $x = R\cos(\omega t - \phi)$:

$$x = R\cos\left(\omega \times \frac{\phi}{\omega} - \phi\right) = R\cos 0 = R.$$

The maximum displacement is R, so this occurs when $t = \frac{\phi}{\omega}$.

14) b

Substitute $t = \frac{\phi}{\omega} + \frac{\pi}{2\omega}$:

$$x = R\cos\left(\omega\left(\frac{\phi}{\omega} + \frac{\pi}{2\omega}\right) - \phi\right) = R\cos\frac{\pi}{2} = 0.$$

So the particle is at equilibrium at $t = \frac{\phi}{\omega} + \frac{\pi}{2\omega}$.

14) c

ϕ is known as the phase constant. It determines the initial position of the particle at $t = 0$ and where in the oscillation the motion begins. Graphically, it translates the graph horizontally.

15) a

$$x = R\cos\omega t, \qquad v = \dot{x} = -R\omega\sin\omega t.$$

15) b

Speed $= |v| = |R\omega\sin\omega t|$. The maximum speed occurs when $|\sin\omega t| = 1$, so the maximum speed is $R\omega$.

15) c

At equilibrium, $x = 0$, so $R\cos\omega t = 0$, giving $\cos\omega t = 0$ and $\sin\omega t = \pm 1$. At these points $|v| = R\omega$. So the maximum speed occurs at equilibrium.

16) a

$$v = \frac{dx}{dt} = \dot{x}, \qquad a = \frac{dv}{dt} = \frac{d}{dt}\left(\frac{dx}{dt}\right) = \frac{d}{dt}(\dot{x}) = \ddot{x}.$$

So $\ddot{x} = \frac{dv}{dt}$.

16) b

Using the chain rule,

$$\frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = \frac{dv}{dx}\,v,$$

so $\ddot{x} = v\frac{dv}{dx}$.

17)

$$\ddot{x} = -\omega^2 x \implies v\frac{dv}{dx} = -\omega^2 x.$$

Separate variables and integrate:

$$\int v\,dv = \int -\omega^2 x\,dx \implies \frac{1}{2}v^2 = -\frac{1}{2}\omega^2 x^2 + k.$$

Therefore $v^2 + \omega^2 x^2 = c$, where $c = 2k$.

18) a

At the furthest points (the turning points), the particle changes direction, so its instantaneous velocity is zero. Hence when $|x| = R$, $v = 0$.

18) b

$$v^2 + \omega^2 x^2 = c.$$

Substitute $x = \pm R$ and $v = 0$:

$$0^2 + \omega^2(\pm R)^2 = c \implies c = \omega^2 R^2.$$

Therefore $v^2 + \omega^2 x^2 = \omega^2 R^2$.

19) a

$$v^2 + 9x^2 = 36 \implies v^2 + 3^2 x^2 = 6^2.$$

This is of the form $v^2 + \omega^2 x^2 = \omega^2 R^2$, so the motion is SHM.

19) b

$\omega^2 = 3^2$, so $\omega = 3$.

19) c

$\omega^2 R^2 = 36$, so $9R^2 = 36$, $R^2 = 4$, and $R = 2$.

19) d

Maximum speed occurs when $x = 0$ (at equilibrium). Then

$$v^2 + 9 \times 0^2 = 36 \implies v^2 = 36 \implies v_{\max} = 6 \text{ m s}^{-1}.$$

20) a

The auxiliary equation is $m^2 + 25 = 0$, so $m = \pm 5i$. Thus $x = A\cos 5t + B\sin 5t$. At $t = 0$, $x = 0.2$, so $A = 0.2$. Also $\dot{x} = -\sin 5t + 5B\cos 5t$. At $t = 0$, $\dot{x} = 0$, so $B = 0$. Hence $x = 0.2\cos 5t$.

20) b

The period is $T = \frac{2\pi}{5}$.

20) c

$$\dot{x} = -\sin 5t,$$

so the maximum speed is 1 m s^{-1}.

20) d

At equilibrium, $0 = 0.2\cos 5t$, so $\cos 5t = 0$. The first time is $5t = \frac{\pi}{2}$, hence $t = \frac{\pi}{10}$ seconds.

21) a

The auxiliary equation is $m^2 + 36 = 0$, so $m = \pm 6i$ and $x = A\cos 6t + B\sin 6t$. The period is $T = \frac{2\pi}{6} = \frac{\pi}{3}$.

21) b

The greatest speed is $\omega R = 6 \times 0.5 = 3$ m s^{-1}.

22) a

The auxiliary equation is $m^2 + 4 = 0$, so $m = \pm 2i$ and $x = A\cos 2t + B\sin 2t$. The period is $T = \frac{2\pi}{2} = \pi$.

22) b

Amplitude $= \frac{v_{\max}}{\omega} = \frac{8}{2} = 4$.

23) a

$$\ddot{x} + \frac{1}{100}x = 0.$$

The auxiliary equation is $m^2 + \frac{1}{100} = 0$, so $m = \pm\frac{1}{10}i$. The general solution is $x = A\cos(t/10) + B\sin(t/10)$ and the angular frequency is $\omega = \frac{1}{10}$.

23) b

$$\ddot{x} = -\frac{1}{100}x \implies v\frac{dv}{dx} = -\frac{1}{100}x \implies \int v\,dv = \int -\frac{1}{100}x\,dx.$$

Thus $\frac{1}{2}v^2 = -\frac{1}{200}x^2 + c$. When $x = 0.3$ and $v = 0.04$,

$$\frac{1}{2}(0.04)^2 = -\frac{1}{200}(0.3)^2 + c \implies c = \frac{1}{800}.$$

So $\frac{1}{2}v^2 = -\frac{1}{200}x^2 + \frac{1}{800}$, and $v^2 + \frac{1}{100}x^2 = \frac{1}{400}$. Hence $\omega^2 R^2 = \frac{1}{400}$ and $\frac{1}{100}R^2 = \frac{1}{400}$, giving $R^2 = \frac{1}{4}$, $R = \frac{1}{2}$. The amplitude is 0.5 m.

23) c

$$T = \frac{2\pi}{1/10} = 20\pi.$$

23) d

Use $v^2 + \frac{1}{100}x^2 = \frac{1}{400}$. When $x = 0.4$,

$$v^2 + \frac{1}{100}(0.4)^2 = \frac{1}{400} \implies v^2 = \frac{9}{10000} \implies v = \frac{3}{100} = 0.03 \text{ m s}^{-1}.$$

24) a

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{\pi/2} = 4.$$

24) b

Use $v^2 + \omega^2 x^2 = \omega^2 R^2$. Substituting $v = 5$, $\omega = 4$, and $x = 0$:

$$5^2 + 4^2 \times 0^2 = 4^2 R^2 \implies 16R^2 = 25 \implies R = \frac{5}{4}.$$

The amplitude is 1.25 m.

24) c

The particle starts from the centre of oscillation, so $x = \frac{5}{4}\sin 4t$. Let $x = 1$:

$$1 = \frac{5}{4}\sin 4t \implies \sin 4t = \frac{4}{5} \implies t = \frac{1}{4}\arcsin\frac{4}{5} = 0.232 \text{ s (3 SF)}.$$

25) a

At equilibrium, $\ddot{x} = 0$. Since $\ddot{x} = -16x + 32$,

$$0 = -16(x - 2),$$

so the equilibrium position is $x = 2$.

25) b

$$\ddot{x} + 16x = 32.$$

For the homogeneous equation, the auxiliary equation $m^2 + 16 = 0$ gives $m = \pm 4i$, so the complementary function is $x = A\cos 4t + B\sin 4t$. Try a constant particular integral $x = k$, so $\dot{x} = 0$ and $\ddot{x} = 0$. Substitution gives $(0) + 16k = 32$, hence $k = 2$. Thus $x = A\cos 4t + B\sin 4t + 2$. Using $x(0) = 3$ gives $A = 1$. Also $\dot{x} = -4\sin 4t + 4B\cos 4t$; using $\dot{x}(0) = 0$ gives $B = 0$. Hence $x = \cos 4t + 2$ and the amplitude is $R = 1$.

25) c

$$T = \frac{2\pi}{4} = \frac{\pi}{2} \text{ s}.$$

26) a

At equilibrium $\ddot{x} = 0$. Since $\ddot{x} = -9x + 27$, $0 = -9(x - 3)$, so the equilibrium position is $x = 3$.

26) b

$$\ddot{x} + 9x = 27.$$

The homogeneous auxiliary equation $m^2 + 9 = 0$ gives $m = \pm 3i$, so the complementary function is $x = A\cos 3t + B\sin 3t$. Try a constant particular integral $x = k$, so $\dot{x} = \ddot{x} = 0$. Substitution gives $9k = 27$, so $k = 3$. Thus $x = A\cos 3t + B\sin 3t + 3$, and $\omega = 3$.

26) c

Substitute $x = 5$ and $t = 0$: $5 = A + 3$, so $A = 2$. Thus $x = 2\cos 3t + B\sin 3t + 3$, and $\dot{x} = -6\sin 3t + 3B\cos 3t$. As the particle starts at $x = 5$ in the positive direction and is moving back towards equilibrium, its velocity is negative. Using $\dot{x} = -3$ at $t = 0$ gives $-3 = 3B$, so $B = -1$. Hence $x = 2\cos 3t - \sin 3t + 3$, and

$$R = \sqrt{2^2 + (-1)^2} = \sqrt{5}.$$

26) d

Maximum speed occurs at equilibrium:

$$v_{\max} = \omega R = 3\sqrt{5} \text{ m s}^{-1}.$$

27)

The period is $T = 8\pi$ and amplitude $R = 1.5$, so $\omega = \frac{2\pi}{T} = \frac{1}{4}$. As the buoy passes upward through equilibrium at $t = 0$, use $x = 1.5\sin(t/4)$; this ensures $x = 0$ and positive velocity at $t = 0$. At A, $x = 0.5$:

$$0.5 = 1.5\sin(t/4) \implies t = 4\arcsin\frac{1}{3}.$$

At B, $x = 1.25$:

$$1.25 = 1.5\sin(t/4) \implies t = 4\arcsin\frac{5}{6}.$$

The time from A to B is

$$4\arcsin\frac{5}{6} - 4\arcsin\frac{1}{3} = 2.58 \text{ s (3 SF)}.$$

28) a

Use $v^2 + \omega^2 x^2 = \omega^2 R^2$. At $x = 0.4$, $v = 1.2$:

$$1.2^2 + \omega^2(0.4)^2 = \omega^2 R^2. \tag{1}$$

At $x = 0.8$, $v = 0.6$:

$$0.6^2 + \omega^2(0.8)^2 = \omega^2 R^2. \tag{2}$$

Subtract (1) from (2): $0.6^2 - 1.2^2 + \omega^2(0.8^2 - 0.4^2) = 0$, so $0.48\omega^2 = 1.08$, $\omega^2 = \frac{9}{4}$, and $\omega = \frac{3}{2}$. Substituting into (1), $1.2^2 + 1.5^2(0.4)^2 = 1.5^2 R^2$, so $R^2 = 0.8$, $R = \frac{2\sqrt{5}}{5}$ m.

28) b

$$T = \frac{2\pi}{3/2} = \frac{4\pi}{3} \text{ s}.$$

29) a

At $x = 15$, $v = 120$: $120^2 + \omega^2(15)^2 = \omega^2 R^2$. At $x = 25$, $v = 80$: $80^2 + \omega^2(25)^2 = \omega^2 R^2$. Subtracting the first equation from the second:

$$80^2 - 120^2 + \omega^2(25^2 - 15^2) = 0 \implies 400\omega^2 = 8000 \implies \omega^2 = 20, \quad \omega = 2\sqrt{5}.$$

Substituting into the first equation gives $R^2 = 945$, so $R = 3\sqrt{105}$ mm.

29) b

$$T = \frac{2\pi}{2\sqrt{5}} = \frac{\pi}{\sqrt{5}} = \frac{\sqrt{5}\pi}{5} \text{ s}.$$

30) a

The force is towards O, so if x is the displacement from O, $F = -18x$. Newton's second law gives $F = ma = 2\ddot{x}$, so $2\ddot{x} = -18x$, hence $\ddot{x} = -9x$.

30) b

This is SHM because it is of the form $\ddot{x} = -\omega^2 x$, with $\omega^2 = 9$, so $\omega = 3$.

30) c

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{3}.$$

31) a

The force is $F = -2x$. Newton's second law gives $F = ma = \frac{1}{2}\ddot{x}$, so $\frac{1}{2}\ddot{x} = -2x$ and $\ddot{x} = -4x$.

31) b

$\omega^2 = 4$, so $\omega = 2$. The general solution is $x = A\cos 2t + B\sin 2t$. At $t = 0$, $x = 0.3$, so $A = 0.3$. Then $\dot{x} = -0.6\sin 2t + 2B\cos 2t$. At $t = 0$, $\dot{x} = 0$, so $B = 0$. Hence $x = 0.3\cos 2t$.

31) c

The first time at O is when $x = 0$: $0 = 0.3\cos 2t$, so $t = \frac{1}{2}\arccos 0 = \frac{1}{2} \times \frac{\pi}{2} = \frac{\pi}{4}$ seconds.

31) d

Use $v^2 + \omega^2 x^2 = \omega^2 R^2$. Substituting $\omega = 2$, $x = 0.2$, and $R = 0.3$:

$$v^2 + 2^2(0.2)^2 = 2^2(0.3)^2 \implies v^2 = \frac{1}{5} \implies v = \frac{\sqrt{5}}{5} \text{ m s}^{-1}.$$

32) a

The force is $F = -12x$. Newton's second law gives $F = ma = 0.8\ddot{x}$, so $\frac{4}{5}\ddot{x} = -12x$, hence $\ddot{x} = -15x$. This is SHM with $\omega^2 = 15$, so $\omega = \sqrt{15}$.

32) b

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{15}}.$$

32) c

$R = 0.25$, so

$$v_{\max} = \omega R = \sqrt{15} \times 0.25 = \frac{\sqrt{15}}{4} \text{ m s}^{-1}.$$

33) a

The restoring force is $F = -8x$. Newton's second law gives $F = ma = 0.5\ddot{x}$, so $\frac{1}{2}\ddot{x} = -8x$ and $\ddot{x} = -16x$. This is SHM with $\omega^2 = 16$, so $\omega = 4$.

33) b

At equilibrium $x = 0$, speed is maximum. $v_{\max} = \omega R$, so $1.2 = 4R$ and $R = 0.3$ m.

33) c

The maximum extension equals the amplitude, so the maximum extension is 0.3 m.

34) a

The general solution of $\ddot{x} = -9x$ is $x = A\cos 3t + B\sin 3t$. At $t = 0$, $x = 0.5$, so $A = 0.5$. Thus $x = 0.5\cos 3t + B\sin 3t$ and $\dot{x} = -1.5\sin 3t + 3B\cos 3t$. Since the box moves towards equilibrium at $t = 0$, $\dot{x} = -1.5$. Hence $-1.5 = 3B$, so $B = -0.5$. Therefore $x = 0.5\cos 3t - 0.5\sin 3t$.

34) b

$$R = \sqrt{0.5^2 + (-0.5)^2} = \frac{\sqrt{2}}{2}.$$

34) c

$$v_{\max} = \omega R = 3 \times \frac{\sqrt{2}}{2} = \frac{3\sqrt{2}}{2} \text{ m s}^{-1}.$$

35) a

The restoring force is $F = -6x$. Newton's second law gives $F = ma = 1.2\ddot{x}$, so $1.2\ddot{x} = -6x$, hence $\ddot{x} = -5x$.

35) b

$\omega = \sqrt{5}$. The general solution is $x = A\cos(\sqrt{5}t) + B\sin(\sqrt{5}t)$. Using $x(0) = 0.6$ gives $A = 0.6$. Differentiating,

$$\dot{x} = -0.6\sqrt{5}\sin(\sqrt{5}t) + B\sqrt{5}\cos(\sqrt{5}t).$$

Using $\dot{x}(0) = 0$ gives $B = 0$. Hence $x = 0.6\cos(\sqrt{5}t)$.

35) c

Equilibrium is when $x = 0$:

$$0 = 0.6\cos(\sqrt{5}t) \implies t = \frac{1}{\sqrt{5}}\arccos 0 = \frac{\pi}{2\sqrt{5}} = \frac{\sqrt{5}\pi}{10} \text{ s}.$$

35) d

Three assumptions: the particle is modelled as a particle; the motion is in a straight line; there is no air resistance.

36) a

The total restoring force is $F = -(5x + 5x) = -10x$. Newton's second law gives $F = ma = 1 \times \ddot{x}$, so $\ddot{x} = -10x$. This is of the form $\ddot{x} = -\omega^2 x$, so the motion is SHM with $\omega = \sqrt{10}$.

36) b

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{10}} = \frac{\sqrt{10}\pi}{5}.$$

36) c

Use $v^2 + \omega^2 x^2 = \omega^2 R^2$. With $\omega = \sqrt{10}$, $x = 0.2$, and $R = 0.4$:

$$v^2 + (\sqrt{10})^2(0.2)^2 = (\sqrt{10})^2(0.4)^2 \implies v^2 = \frac{6}{5} \implies v = \frac{\sqrt{30}}{5} \text{ m s}^{-1}.$$

37) a

The total restoring force is $F = -(4x + 4x) = -8x$. Newton's second law gives $F = ma = 2\ddot{x}$, so $2\ddot{x} = -8x$ and $\ddot{x} = -4x$. This is SHM with $\omega^2 = 4$, so $\omega = 2$.

37) b

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi.$$

37) c

x is the distance of the package from O towards B. If the package is 1.1 m from A, its displacement is $x = 1.1 - 1.5 = -0.4$ m. Use $v^2 + \omega^2 x^2 = \omega^2 R^2$ with $\omega = 2$, $x = -0.4$, and $R = 0.6$:

$$v^2 + 2^2(-0.4)^2 = 2^2(0.6)^2 \implies v^2 = \frac{4}{5} \implies v = \frac{2\sqrt{5}}{5} \text{ m s}^{-1}.$$

38)

The total restoring force is $F = -(9mx + 9mx) = -18mx$. Newton's second law gives $F = ma = m\ddot{x}$, so $m\ddot{x} = -18mx$ and $\ddot{x} = -18x$. This is SHM with $\omega^2 = 18$, so $\omega = 3\sqrt{2}$. The puck starts at O, so

when $t = 0$, $x = 0$ and, since it is tapped towards A with speed 0.8, $\dot{x} = 0.8$. The greatest speed occurs at equilibrium ($x = 0$), so the initial speed is maximum: $v_{\max} = 0.8$. Thus

$$v_{\max} = \omega R \implies 3\sqrt{2}R = 0.8 \implies R = \frac{2\sqrt{2}}{15}.$$

As $OA = 1$ m, the puck is closest to A at the turning point, so the distance is $1 - \frac{2\sqrt{2}}{15} = 0.811$ m (3 SF).

39) a

At equilibrium, net force is 0 N. Take positive direction towards B:

$$-10mx + 5m(2.4 - x) = 0 \implies -2x + (2.4 - x) = 0 \implies 3x = 2.4 \implies x = 0.8.$$

So the equilibrium position is 0.8 m from A.

39) b

The total restoring force is

$$F = -10mx + 5m(2.4 - x) = -10mx + 12m - 5mx = 12m - 15mx.$$

Newton's second law gives $F = ma = m\ddot{x}$, so $m\ddot{x} = 12m - 15mx$, hence $\ddot{x} = 12 - 15x = -15(x - 0.8)$. This is SHM about $x = 0.8$ with $\omega^2 = 15$, so $\omega = \sqrt{15}$.

39) c

The right-hand spring force is $5m(2.4 - x)$. This is largest when $2.4 - x$ is largest, so when x is as small as possible during the motion. The trolley is released from rest 0.3 m to the right of equilibrium, so this is at $x = 0.8 + 0.3 = 1.1$ m. As it is released from rest, this point is also a turning point, and the amplitude is $R = 0.3$. The two extreme positions are $x = 0.8 \pm 0.3 = 1.1$ or 0.5. The greatest right-spring force occurs at $x = 0.5$:

$$F = 5m(2.4 - 0.5) = \frac{19}{2}m \text{ N}.$$